

Joseph Flammia

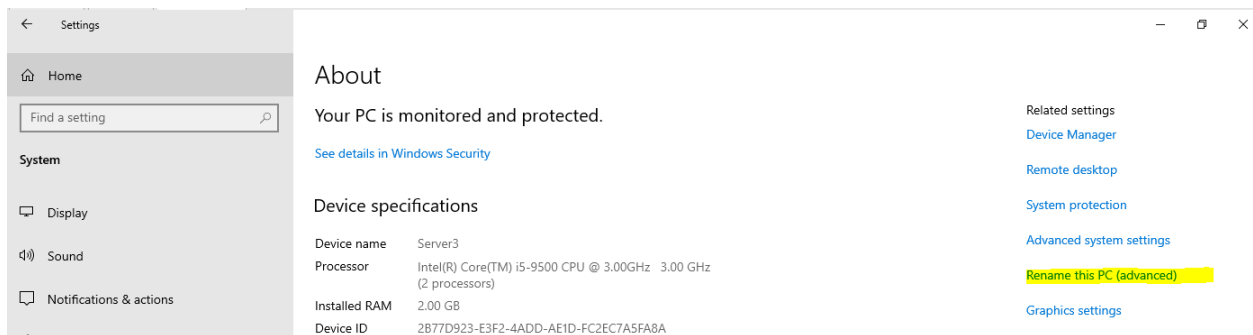
Lab 11

3/5/25

CYB126

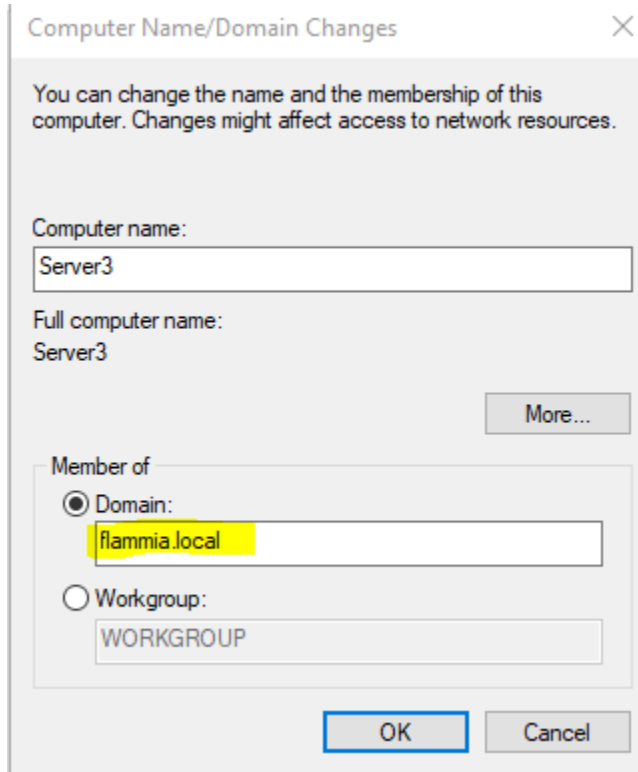
Lab 11A:

To start off, we will be adding our Windows 3 Server to our domain. This process will be nearly identical to Windows 11. Simply navigate to the taskbar in the Windows 3 Server, open settings, and navigate to the “About” page. From there, click on “Rename this PC (advanced)” on the right-hand side.

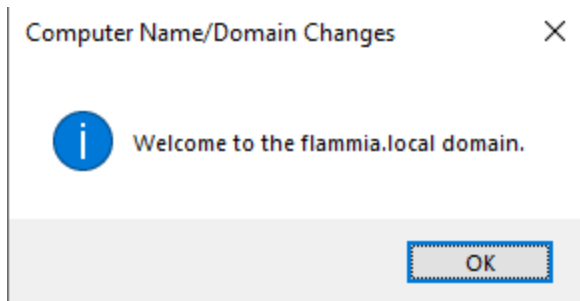


If done correctly, we will be prompted with a screen to either change computer name or enter our system into a domain. Here, I entered the name of my domain, “flammia”.

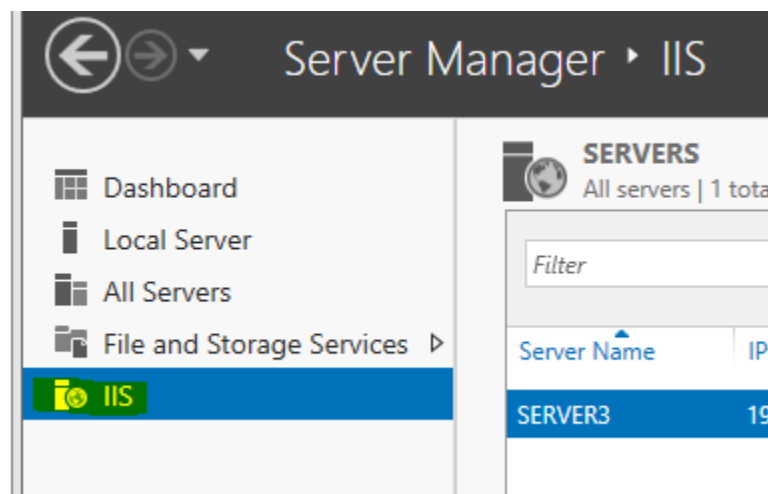
NOTE: Ensure that in your address properties, your preferred DNS server is pointing towards the IP of the desired domain. Otherwise, you will be unable to find the domain and greeted with an error message.



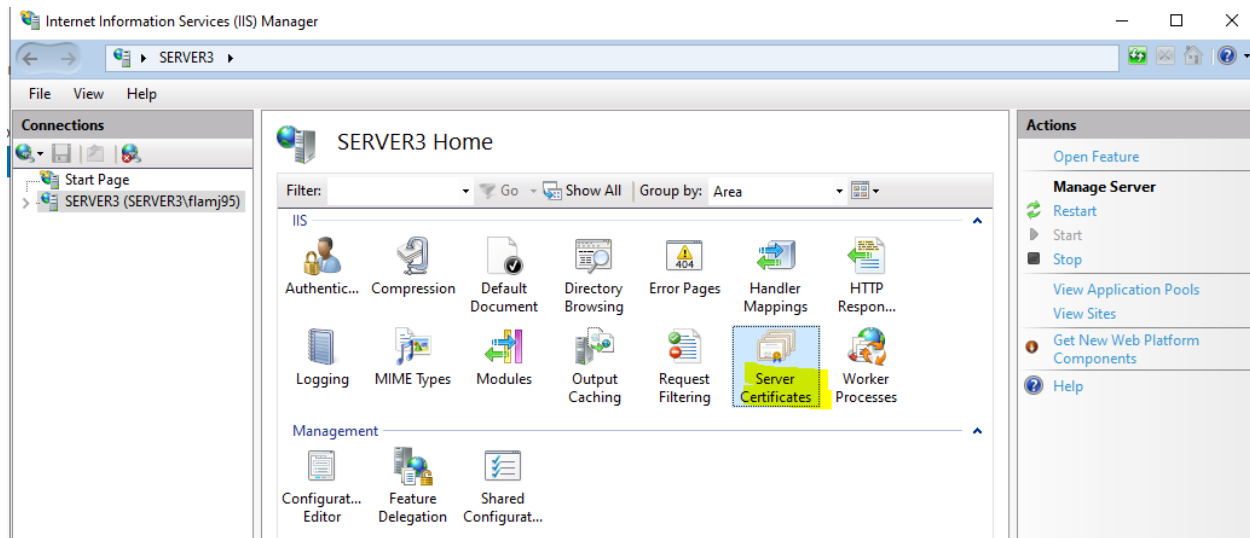
After entering an administrative username and password, I was able to connect server 3 to my domain, “flammia.local”.



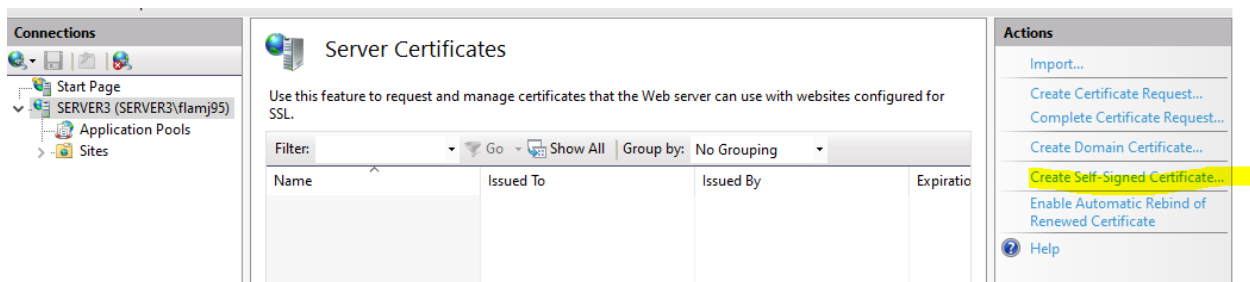
For this next step, we will have to make sure IIS is installed on our Windows 3 Server. To do this, simply make sure that IIS is an option in the “Server Manager” application.



Once verifying IIS is successfully installed, we can open the IIS manager by searching for it in the taskbar on our Windows Server. Once opened, we will navigate to the “Server Certificates” option to create a self-signed certificate for our server.



Once opened, we will click the “Self Signed” option, found on the right-hand side of the application.



From here, we will be prompted to enter a name for our certificate. Here, I entered the name “flammiacert”. The certificate store part can be left as the default for this portion.

Create Self-Signed Certificate ? X

 **Specify Friendly Name**

Specify a file name for the certificate request. This information can be sent to a certificate authority for signing:

Specify a friendly name for the certificate:

flammiacert

Select a certificate store for the new certificate:

Personal

OK Cancel

If done correctly, we can now see that the certificate has been created with some basic information about the certificate.

 **Server Certificates**

Use this feature to request and manage certificates that the Web server can use with websites configured for SSL.

Filter:	Go	Show All	Group by: No Grouping
Name	Issued To	Issued By	Expiration Date
flammiacert	Server3.flammia.local	Server3.flammia.local	3/4/2026 7:00:00 P

From here, we will need to configure our web server to use the certificate. To do this, click “bindings” on the right-hand side as shown below.



Sites

Filter:

Go

Group by: No Grouping

Name	ID	Status	Binding	Path
 Default Web Site	1	Started (ht...	*:80 (http)	%SystemDrive%\inetpub\ww

Actions

 Add Website...

 Set Website Defaults...

Edit Site

 Bindings...

 Basic Settings...

 Explore

 Edit Permissions...

 Remove

 Rename

 View Applications

 View Virtual Directories

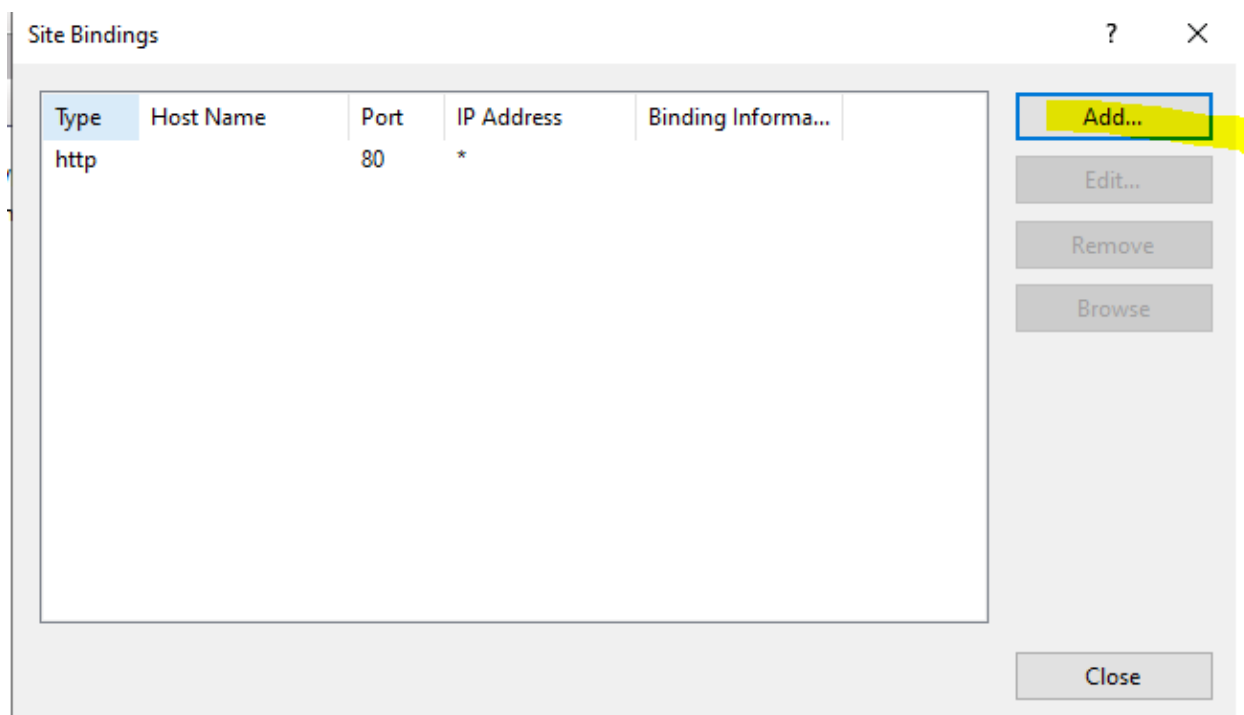
Manage Website

 Restart

 Start

 Stop

Remove Website



Next, we will configure the “https” type. As shown below, the only thing we need to do in this step is enable the “SSL certificate”. This will enable HTTPS on our webserver.

Edit Site Binding ? ×

Type:	IP address:	Port:
https	All Unassigned	443

Host name:

☐ Require Server Name Indication

☐ Disable TLS 1.3 over TCP ☐ Disable QUIC

☐ Disable Legacy TLS ☐ Disable HTTP/2

☐ Disable OCSP Stapling

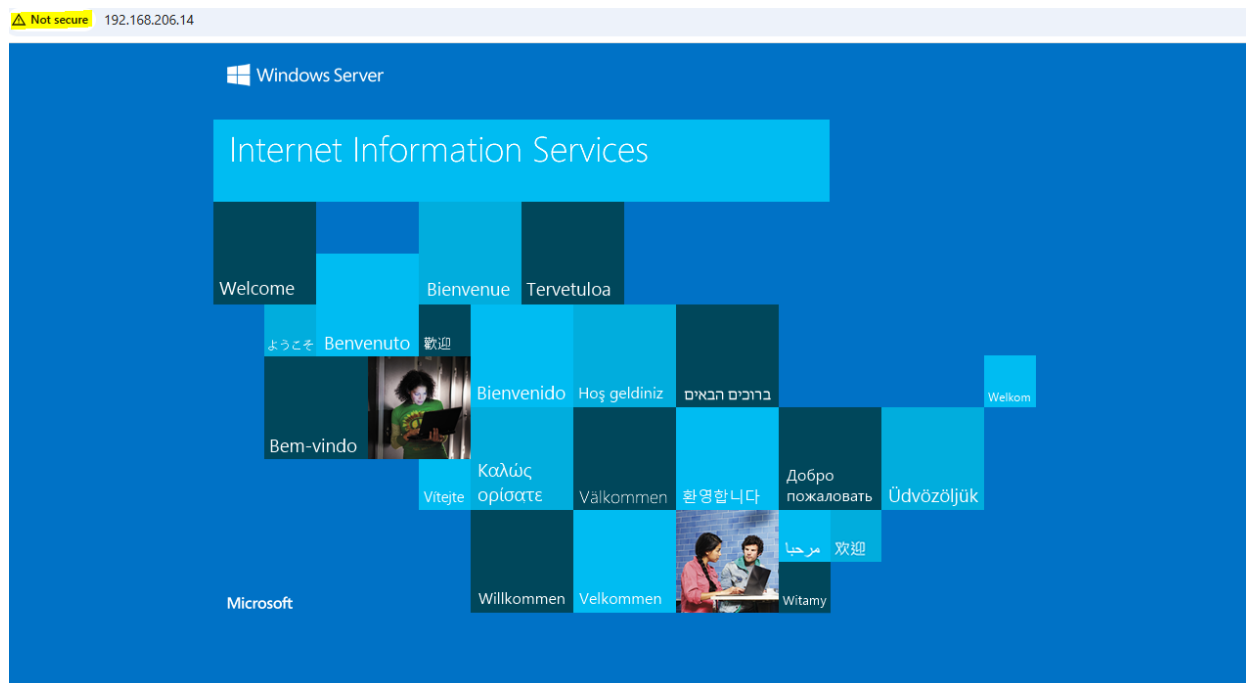
SSL certificate:

flammiacert

Select... View...

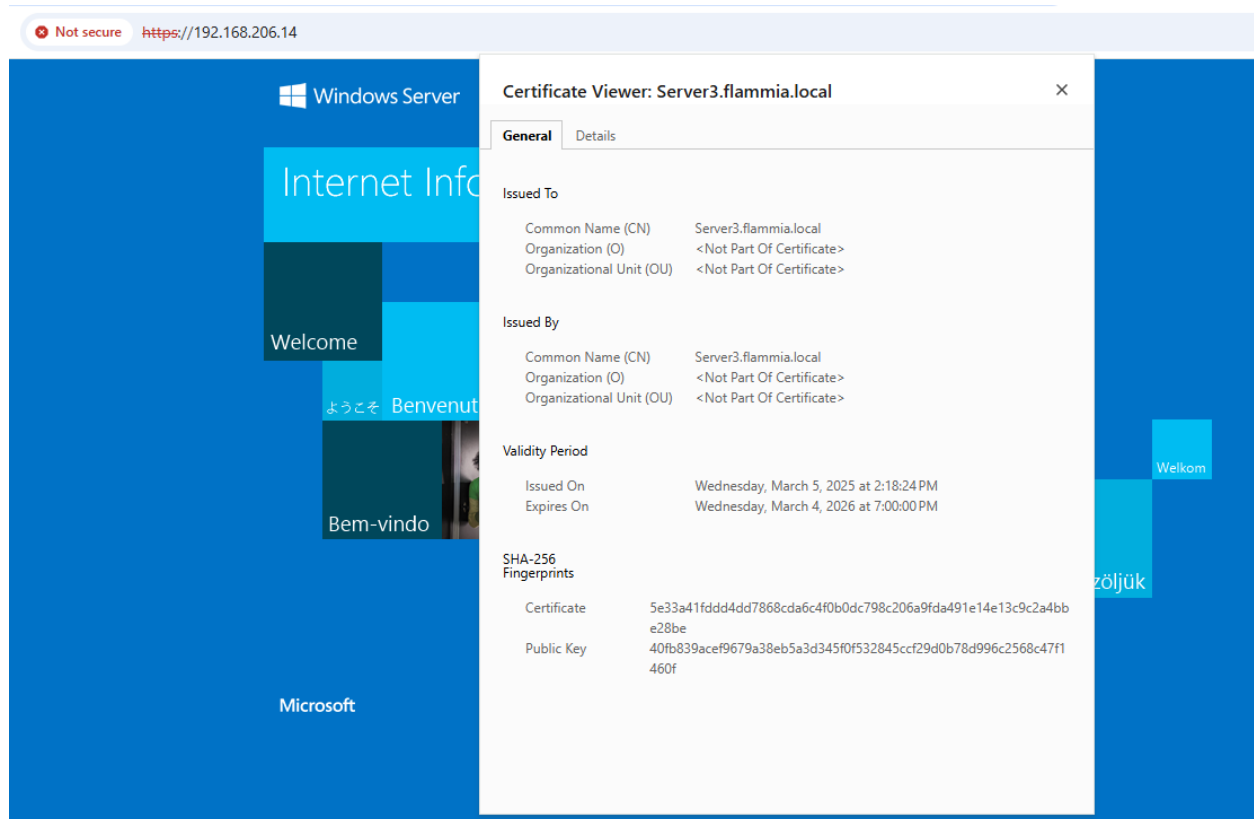
OK Cancel

Now to test if our configurations are correct, I will attempt to reach the web servers from my classroom computer. Now to test the “HTTP” type. Here, I entered “<https://192.168.206.14>” into my web browser and was greeted with the IIS webpage. However, because it is an “http” type, it is considered not secure.



Now to test the “HTTPS” type. Here, I entered “<https://192.168.206.14>” into my web browser and was greeted with the IIS webpage. However, it is still considered insecure due to the self-signed certificate. This is because the certificate does not come from a trusted CA such as GoDaddy or GlobalSign.

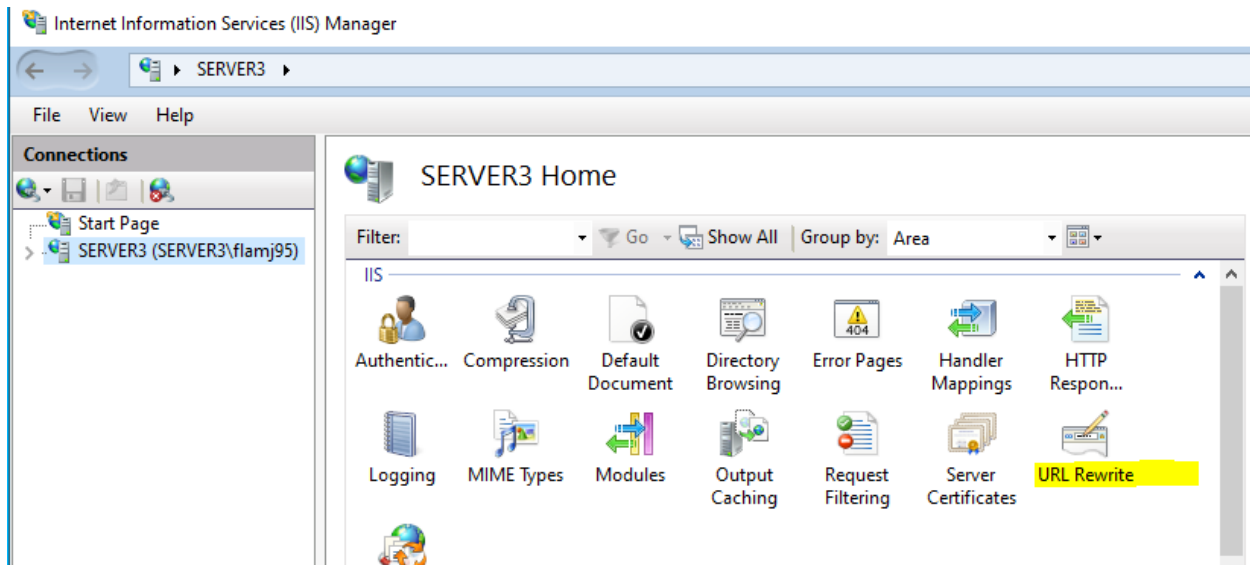
Additionally, we can view some information about the certificate by clicking “Not Secure” and then “Certificate Details”. Here, we can view information such as the name of the server the website is hosted on, and the date the certificate was issued.



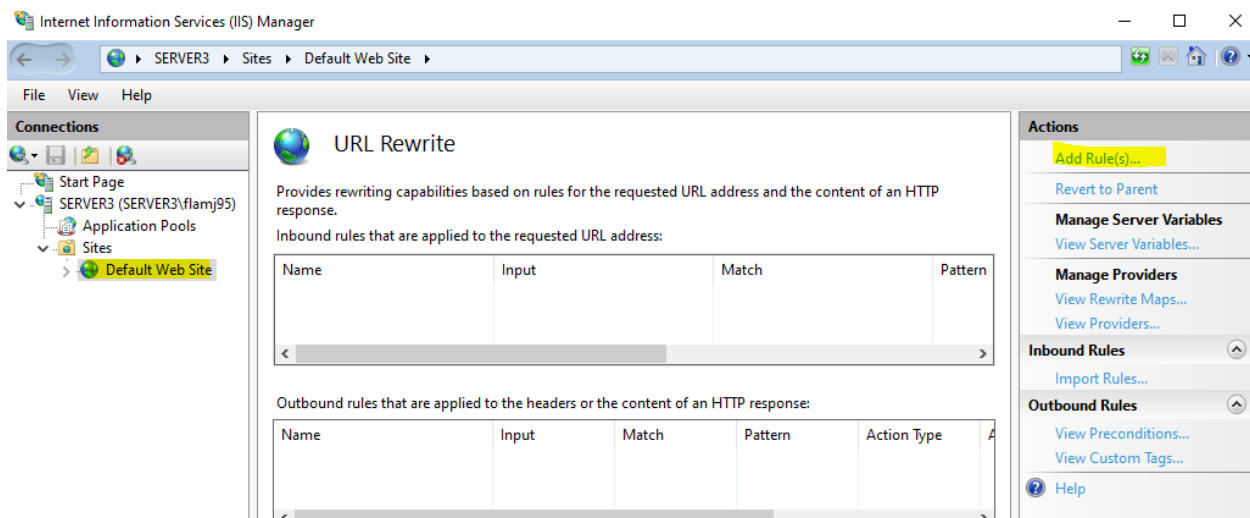
Lab 11B:

In this portion of the lab, we will be installing “URL Rewrite” on our IIS system. To start off, we will install it on our host computer and then drag and drop it onto our Windows Server (IIS Server).

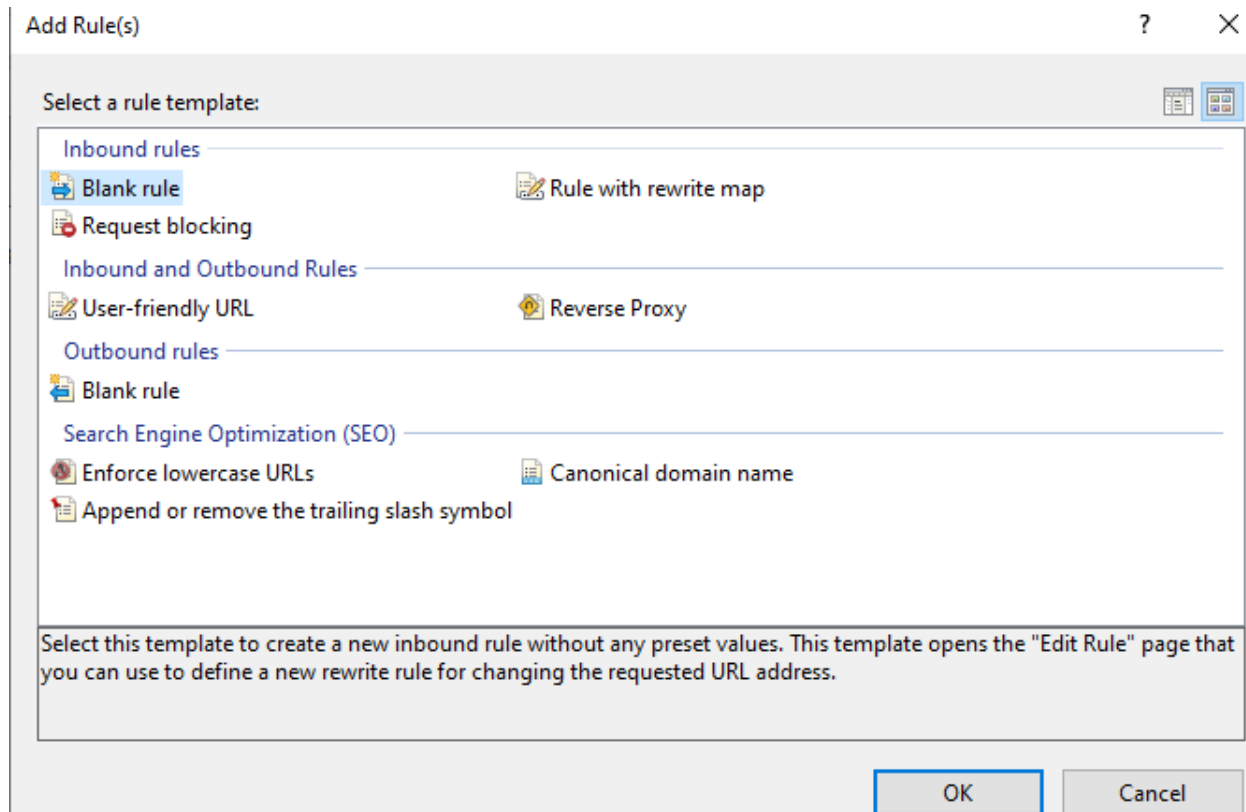
After installing, navigate to the IIS manager and look for the “URL Rewrite” option. This option will only appear after installing the URL Rewrite file.



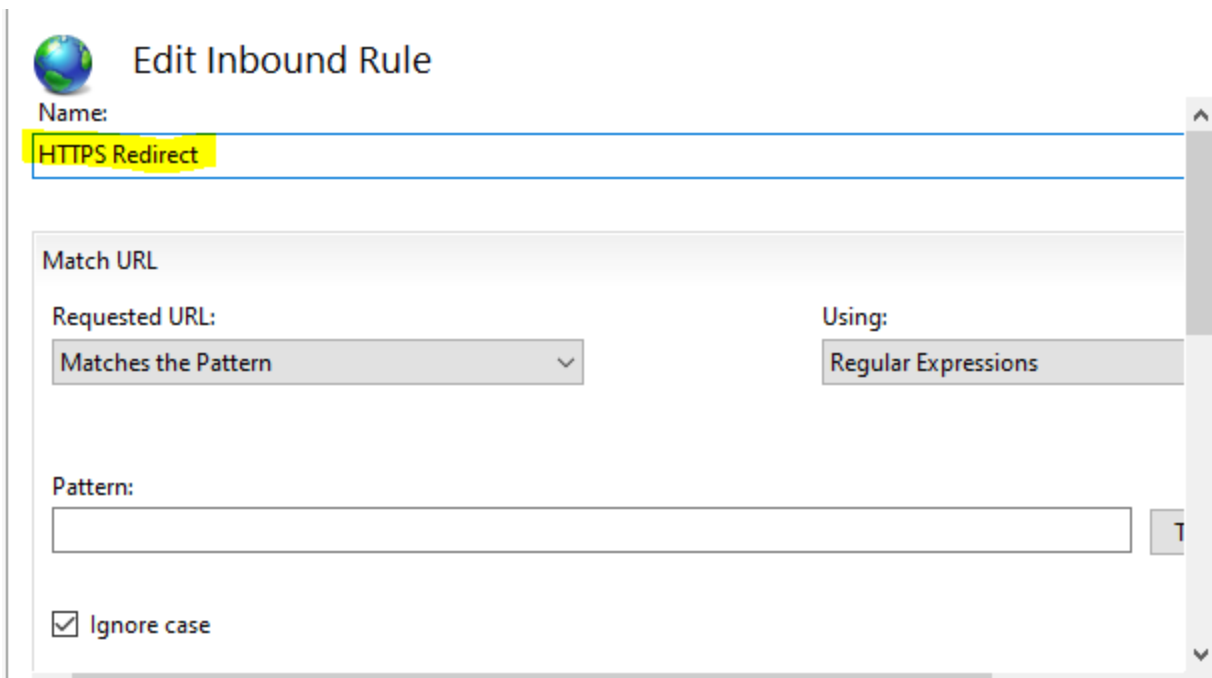
From here, click on the desired website you wish to use and then click “Add Rule” on the right-hand side of the application.



From here, we can choose the kind of rule we would like to add. For now, just click on “Blank Rule”.



The next few steps are important to remember. To start off, name the rule something straightforward and simple to remember. Here, I named the rule “HTTPS Redirect”.



 Edit Inbound Rule

Name:
HTTPS Redirect

Match URL

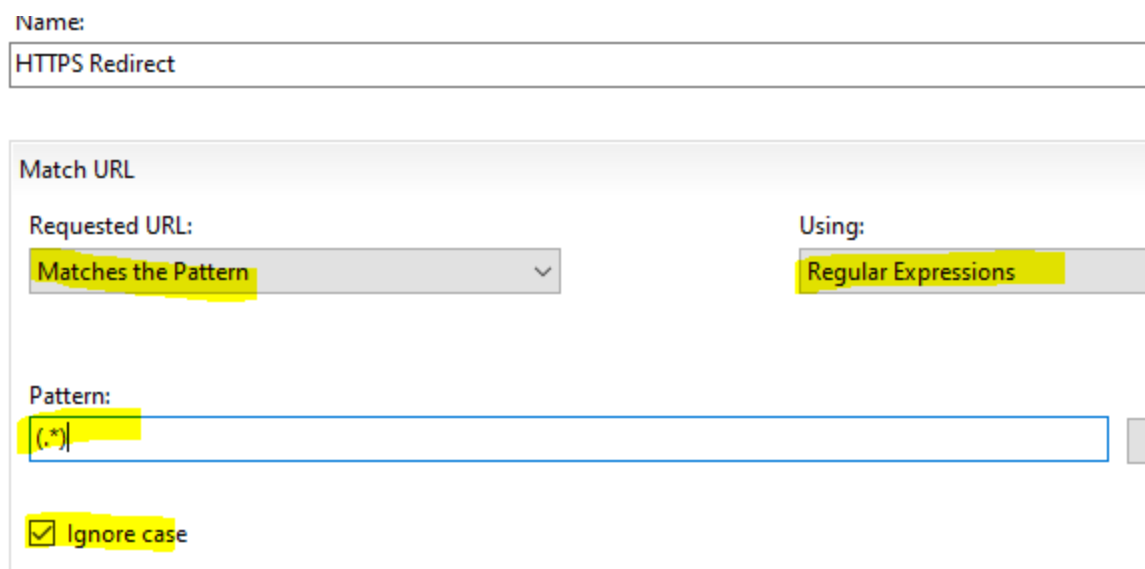
Requested URL:
Matches the Pattern

Using:
Regular Expressions

Pattern:

☒ Ignore case

Now located right below, make sure that the requested URL option is set to “Matches the Pattern”, is using “Regular Expressions”, the pattern used is “(.*)”, and the “Ignore Case” box is checked off.



Name:
HTTPS Redirect

Match URL

Requested URL:
Matches the Pattern

Using:
Regular Expressions

Pattern:
(.*)

☒ Ignore case

Next up, scroll down and double click on the “conditions” section. From here make sure that the “Logical grouping” option is set to “Match All”, and then click Add.



Edit Inbound Rule

Conditions

Logical grouping:
Match All

Input	Type	Pattern

Add...

Edit...

Remove

Move Up

Move Down

Clicking add will open a dialog box. In this box, the following option should be set. For condition input, enter {HTTPS}, check if input string should be set to “Matches the Pattern”. Below that, the pattern should be “^OFF\$” and finally make sure that the “Ignore Case” box is checked off.

Add Condition

Condition input:
{HTTPS}

Check if input string:
Matches the Pattern

Pattern:
^OFF\$

Test pattern...

☒ Ignore case

OK

Cancel

After pressing “OK”, we should see our condition in the “conditions” list.



Edit Inbound Rule

Conditions

Logical grouping:
Match All

Input	Type	Pattern
{HTTPS}	Matches the Pattern	^OFF\$

Add...

Edit...

Remove

Move Up

Move Down

Now in the “Actions” tab located below, make sure the options match the picture below:

Action

Action type:
Redirect

Action Properties

Redirect URL:
https://{HTTP_HOST}/{REQUEST_URI}

☐ Append query string

Redirect type:
Permanent (301)

If everything checks out, we can hit “apply” on the right-hand side of the application to save our changes.

Actions

✓ Apply

✗ Cancel

← Back to Rules

ⓘ Help

To verify our configurations were correct, we can check our redirection in a browser on our host computer.

As you can see, the “https” part is automatically added to the url. If I were to try and type <http://192.168.206.14/>, the url would automatically redirect to the https one instead.

